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Expanding retrieval practice: An effective aid to preschool children's learning

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The benefits of expanding retrieval practice for preschool children were explored in two experiments. In Experiment 1, three groups learned names for six plush toy pigs using expanding retrieval practice, a reward incentive, or a control condition. Reward did not significantly improve learning but retrieval practice doubled recall. In Experiment 2, three groups learned names to soft toys, comparing recall following massed elaborative study with either expanding retrieval practice or expanding re-presentation. Recall was tested after 1 minute, 1 day, and 2 days. A very large effect size ($d = 1.9$) indicated the very considerable benefit from expanding retrieval practice over the elaboration condition. Comparison with the re-presentation condition suggested that half of the benefit of expanding retrieval practice came from spaced scheduling and half from retrieval practice. Expanding retrieval practice provides an effective method to improve learning by young children.

Preschool children can develop a very rich and detailed knowledge of topics that interest them, such as dinosaurs (e.g., Chi & Koeske, 1983). However, in general, their memory for new experiences is poorer than that of older children. For example, they have poor memory of objects that they have been shown (e.g., Perlmutter & Myers, 1979), and they forget more quickly than do older children (e.g., Brainerd & Reyna, 1995). One reason for the poorer memory performance of preschool children is their ignorance of what is likely to influence their memory. For example, preschool children are unaware of retroactive interference and of the vulnerability of their memories to suggestion, although by Grade 3 children are aware of these influences (e.g., O'Sullivan, Howe, & Marche, 1996). Another difference between preschool and older children is that preschool children generally do not adopt strategies to help them learn (e.g., Flavell, Friedrichs, & Hoyt, 1970; Perlmutter & Myers, 1979). Preschool children are also very poor at predicting their memory performance (Flavell et al., 1970). Other reasons for the poorer learning of targeted

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information (i.e., information that teachers may be trying to teach the preschool child) are young children's greater impulsivity, their vulnerability to distraction, and their susceptibility to task interference (e.g., Pressley, 1979).

Given the relatively poor learning and retention by preschool children, combined with their problems with sustained attention and their limited metamemory knowledge and skills, we believed that there was a strong case for exploring with preschool children the effectiveness of a simple memory improvement technique. Spaced retrieval practice is a robust and powerful technique for improving learning that has benefited learning in many situations for normal adults (Cull, 2000; Cull, Shaughnessy, & Zechmeister, 1996; Landauer & Bjork, 1978; Morris & Fritz, 2000, 2002; Morris, Fritz, & Buck, 2004; Morris, Fritz, Jackson, Nichol, & Roberts, 2005), 8-year-old schoolchildren (Rea & Modigliani, 1985), the elderly suffering dementia (Camp, Foss, O'Hanlon, & Stevens, 1996), and older adults who have suffered the consequences of HIV infection (Lee & Camp, 2001). However, the technique has not been tested with children younger than 8 years old, so from an applied standpoint, it would seem reasonable to test its effectiveness in helping very young children to learn.

In addition, the impulsive and strategically unsophisticated nature of preschool children makes them a very suitable population for investigating the degree to which the effects of the retrieval practice technique are automatic (i.e., nonwilful). Participants in other experiments involving retrieval practice (e.g., Fritz, Morris, Acton, Voelkel, & Etkind, in press) often appear to be somewhat surprised at having learned the practised material even when they had not expected to learn it and so had not been especially dedicated to the task. This sort of report, combined with the almost universal benefits of retrieval practice, suggest that improvement in later memory performance may occur as an automatic outcome of well-scheduled retrieval practice. If preschool children show a pattern of benefits that is similar to that of adults and older children, then it seems likely that the benefits are at least somewhat automatic.

Expanding retrieval practice is a specific type of spaced practice that seems quite suitable for learning among preschool children because it simply involves attempting to retrieve items on a few occasions: first just after presenting the information, then following gradually increasing delays. For example, Morris et al. (2005) studied the learning of names to pictures of faces by secondary-school students. The pairing of each face with its name was shown just once to the students but was followed by three practice tests that asked students to recall the name when the face was reshown. The retrieval practice in this study had a 0–3–6 schedule: The first occurred immediately following the presentation; the second test followed the first after three intervening events; then the third practice test followed six more events. In a subsequent criterion test the students who had followed this schedule of expanding retrieval practice recalled more than twice as many names as did students who were repeatedly shown the face–name pairings for study on the same schedule, but without practice tests.

Theoretical accounts of the effectiveness of retrieval practice have emphasized the importance of desirable difficulty in the recall attempt. For retrieval practice to benefit learning, recall must be sufficiently difficult but not so difficult that it cannot be accomplished (Bjork, 1988; Bjork & Bjork, 1992). Because each retrieval increases the likelihood that the material can be retrieved in the future, the spacing between retrieval attempts is increased to maintain a more balanced degree of difficulty across trials; in contrast, a schedule with uniformly short lags would allow for early success, but the following practice would be trivial because it would be too easy, whereas a schedule with uniformly long lags would not enable early success. As well as very effectively improving learning, retrieval practice makes the material learned more resistant to subsequent forgetting (Wheeler, Ewers, & Buonanno, 2003).

Given the research reviewed, this research asks whether the benefits of expanding retrieval practice can be extended to learning by preschool children. This question extends previous research on the technique, specifically considering whether

the benefits observed in learning proper names (e.g., Morris et al., 2005) extend to very young children who have not yet developed strategic memory skills.

EXPERIMENT 1

In Experiment 1 preschool children, tested individually, learned the names of plush toy pigs using one of three methods. In the comparison condition the children were carefully introduced to each of the toys, and the child said the name of each toy. When all the toys had been introduced children were tested on the names. The reward condition was similar, except that the children were told in advance that they would be given a sticker if they tried their best. In the expanding retrieval practice condition the introduction of the toy pigs was interspersed with requests for their names following an expanding schedule. We predicted that the reward condition would improve performance over the comparison condition but that, based upon previous research on retrieval practice, the expanding retrieval practice condition would lead to the best recall.

Three preschools were involved, with each preschool assigned to a particular study method. Assigning only one study method to each preschool prevented the children from discussing and comparing the various study methods. However, it introduced the possibility that the schools as well as the individual children might differ. A baseline measure of name learning under controlled conditions was included to identify preexperimental group differences and to enable an improvement score to be derived for each child.

Method

Participants

A total of 60 preschool children, 33 female and 27 male, aged between 2.5 and 5 years, were tested. The children regularly attended one of three preschool playgroups; 20 children were tested from each playgroup. Letters were sent to parents informing them of the study and of their right to withdraw their child. Children were also given the opportunity to leave at any point during the study, but none chose to do so.

Design

Three study methods were compared across participant groups: Children were introduced to plush toy pigs with no other instructions (to provide a baseline as a control condition), with a reward-based incentive, or with expanding retrieval practice. Each school was randomly assigned one of the study methods. In the experimental sessions, each child studied the names of six plush toy pigs. Recall of the names of the plush toys was tested immediately after the study session ended. In addition to the measure of performance from the experimental session, for every child a baseline measure of name learning was made during the week before the experimental sessions. For this measure, all children followed the control condition procedure as they attempted to learn the names of a different set of plush toys (bears).

Materials

Pilot research indicated that 6 toys were an appropriate number for the initial baseline and the experimental conditions. Therefore, 12 plush toy animals were used: 6 teddy bears for the initial baseline and 6 pigs in the experimental sessions. The same toys were used for every child. The two different types of animal were used to avoid possible confusion between sets. Each toy was assigned a name: either a human name (e.g., Charlie) or an animal name (e.g., Tinker). None of the toys represented specific, well-known characters (e.g., Winnie the Pooh, Porky Pig), and none of the names was associated with any obvious feature of a toy (e.g., Spotty). The order of the toys (and names) was randomly determined once, and that order was used for all children. In addition to the toys, stickers were used in one study method as a reward incentive.

Procedure

General. Each child was instructed and tested individually in a quiet area away from other children.

First, a brief conversation took place between the experimenter and the child to put the child at ease. After each child finished a session, he or she was praised for making an effort and then rejoined a play group.

Initial baseline session. Each preschool was visited a week before the experimental session. Individually, children were informed that they would be introduced to "some friends" and that they should try to learn as many of their names as possible. Six teddy bears were then introduced using standardized instructions. The teddies were kept in a bag and were pulled out one at a time. The child was encouraged to say "hello" to the teddy using its name and to touch the teddy by shaking its hand or hugging it. The teddy was then placed on a table in front of the child, and the child was encouraged to say "goodbye" to the teddy, again using the name. After all six teddies had been introduced, the child's memory for the toys' names was tested. A standard test procedure was used throughout the experiment: First the child was asked to name any of the toys that he or she could remember; then the experimenter pointed to the toys in order and specifically asked the child to provide the name for any toy not previously named spontaneously. Each child was given praise regardless of performance.

Experimental session. Each preschool was visited the following week, on the same day and at the same time, having been assigned one of the three study methods. Six plush pigs, with different names from those of the teddies, were used.

For the control condition the procedure was exactly the same as that for the initial baseline session. For the reward-based study method, the standardized instructions were supplemented with the promise of a reward for best effort: At the beginning of the task children were informed that they would be given a sticker if they "tried their best". At the end of the session each child received a sticker. Although children were told that the reward of a sticker was conditional upon effort, the stickers were always awarded because otherwise children who did not get stickers might have been distressed through their participation in the study.

For the expanding retrieval practice method, the procedure followed the schedule described in Table 1. *Introduction* involved the experimenter naming the pig and the child saying "hello" to the toy. A *revisit* required the child to recall the name. If the pig was named, the experimenter said "well done". If the child recalled the name incorrectly, or could not name the pig, the experimenter repeated the name and encouraged the child to say it. For each toy (row in Table 1), the lags increased between subsequent events: For the first toy, lags were 1, then 2, then 3, then 7 events (1–2–3–7), escalating the demands on the child's memory. In the interests of brevity, later toys were revisited less frequently.

Results

The .05 significance level was adopted for all significance tests reported in this paper. The number of names correctly recalled was calculated for each child and was aggregated for each of the three groups in the baseline and the experimental sessions. The means and standard deviations for the study conditions in the two sessions are shown in Table 2. Performance in the baseline session was analysed first to establish whether there were differences between the ability of the children in the three schools that needed to be taken into account when analysing the experimental session. A one-way analysis of variance (ANOVA) was carried out on the baseline data; it identified a significant effect of group, $F(2, 57) = 6.47$, $MSE = 1.15$, $p < .01$, indicating initial significant differences among the groups.

To compensate for the differences revealed by the baseline analysis, an analysis of covariance (ANCOVA) was conducted with experimental study condition (control, reward, and retrieval practice) as the main variable and baseline performance as the covariant. This analysis revealed a significant difference between the experimental study conditions, $F(2, 56) = 26.45$, $MSE = 1.06$, $p < .01$, as well as a significant effect of baseline performance, $F(1, 56) = 18.09$, $MSE = 1.06$,

Table 1. *The schedule followed in the expanding retrieval practice condition of Experiment 1*

	Event sequence																									
Toy	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*16*	*17*	*18*	*19*	*20*	*21*	*22*	*23*	*24*	*25*	*26*
Pinkie	I		R			R				R								R								
Rolo		I		R			R						R							R						
Bella					I			R			R				R								R			
Percy									I			R				R								R		
Tinker														I			R				R				R	
Freddie																			I			R				R

Note: I = the introduction of a toy's name. R = a revisit to the toy involving retrieving the name.

Table 2. *Means for recall of the names of the toys in the baseline and experimental tests of Experiment 1*

	Mean recall[a]		
	Control	*Reward*	*Retrieval practice*
Baseline	1.90 (1.02)	1.10 (1.02)	2.30 (1.17)
Experimental	1.95 (1.23)	1.85 (1.31)	4.45 (0.95)

Note: Standard deviations in parentheses.
[a]Maximum possible = 6.

$p < .01$. The corrected means from the ANCOVA for the three study conditions are shown in Figure 1. A Fisher–Hayter analysis (advised by Keppel & Wickens, 2004) was carried out to test the significance of the differences between pairs of study conditions. Recall in the retrieval practice condition was significantly better ($p < .01$) than in either the control or the reward condition, and the last two conditions did not differ significantly. Cohen's *d* indicated very large effect sizes for the corrected differences between the retrieval practice condition and the reward condition ($d = 1.8$) and between retrieval practice and the control condition ($d = 2.2$). The effect size for the difference between the reward and the control condition was small, $d = 0.3$. Figure 2 displays the distributions for the three experimental conditions; for each child, performance in the baseline test has been subtracted from correct name recall in the experimental test. These histograms illustrate that no child performed worse in the retrieval practice than in the baseline condition and that almost all of the children in the retrieval practice condition benefited from the technique.

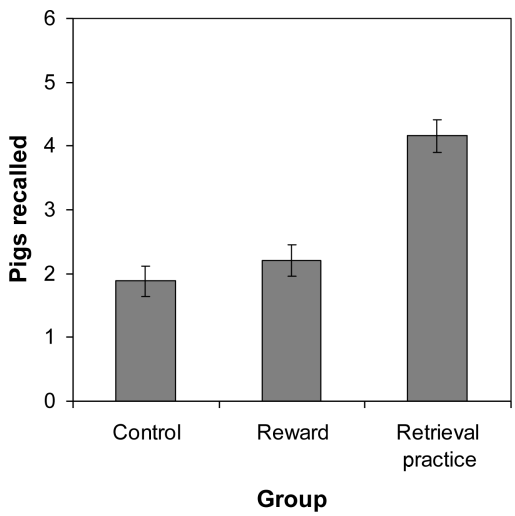


Figure 1. *Experiment 1: Mean recall of the names of the toy pigs, corrected for baseline performance. Error bars indicate* ± *1* SE.

Discussion

Although differences between the preschools on the initial performance of the children were confirmed by the analysis of the baseline recall, the analysis of covariance allowed us to control for the initial ability of the children, both between and within the three schools. With this initial difference controlled it was possible to evaluate the effect of expanding retrieval practice and the offer of reward for effort. The effect of expanding retrieval practice upon children's cued recall performance was dramatic. The children in the retrieval practice condition recalled more than twice as many names as did the children in the control condition. Expanding retrieval practice was also much more effective than the promise of a reward. The reward condition was not significantly better for learning the names than the control condition; the observed effect was small.

EXPERIMENT 2

In Experiment 1 we demonstrated powerful benefits of expanding retrieval practice for preschool children learning the names of plush toys. The children using expanding retrieval practice were able to remember more than twice as many names as could the children in the control condition. However, a confound in Experiment 1 limits clear interpretation of the results. In the expanding retrieval practice condition children typically spent longer interacting with the toys and heard the names more often. The focus of our research concerns the benefits of retrieval

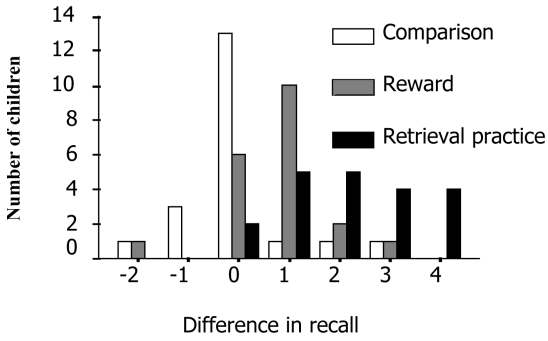


Figure 2. *Experiment 1: Distributions of children using the three study methods, showing the difference between their experimental and baseline scores.*

practice, so Experiment 2 removed this confound, isolating and examining the effects of spacing and retrieval.

In Experiment 2 three experimental conditions were tested: In addition to the expanding retrieval practice condition there was a matched re-presentation condition and a massed elaboration condition. In both of these new conditions the amount of time that the children spent with each of the toys and the number of times that they heard and said the toy's name were matched to the retrieval practice condition. In the re-presentation condition the names of the toys were repeated by the experimenter and elicited from the child following the same schedule as that used for expanding retrieval practice. In this way, the influence of the attention given to the toy, the number of times that the name was heard by the child in association with the toy, and the effect of the spaced schedule for attending to the toy were matched with the expanding retrieval practice condition. Any observed difference between the retrieval practice and the re-presentation conditions can therefore be ascribed to practice retrieving the names. In the massed elaboration condition, total time was evenly distributed among the toys, with one extended discussion of each toy. The discussion began with an introduction and remained focused on the target toy with the experimenter saying the name and eliciting the name from the child as often as in the other conditions. Thus, any difference between the massed elaboration and the re-presentation conditions can be attributed to the benefits of the spaced reencounters of the names.

A second purpose of Experiment 2 was to examine longer term memory for the names. In Experiment 1 memory for the names was tested shortly after learning. In Experiment 2 we added a delayed test after 24 hours. Possible interference effects were also examined by teaching the preschool children the names to a new set of animals on the following day. This allowed an evaluation of the effectiveness of the retrieval practice method beyond the effects of its first use and consideration of possible interference deriving from substantial new or old learning.

Method

Design

Three study methods were compared among participants: expanding retrieval practice, expanding re-presentations, and massed elaboration. In addition, a baseline measure of children's name-learning performance was obtained to test for a priori differences between children assigned to the different study methods. In the experimental conditions, each child studied the names of two different sets of seven plush toys, one per day on two different days. The toy set used in each session was counterbalanced so that for half of the children in each study condition one set was used first, and for the other half the other set was used first. Each set of toy-name pairs was tested 1 minute after study and again after 1 day. The set used in the first experimental session was also tested at the end of the final session—that is, 2 days after study. Although each set contained seven toys, performance on the first toy was not included in the analysis to avoid primacy effects. Also, the first toy appeared more frequently in the study phase to enable appropriate spacing in the schedule.

Participants

A total of 62 children (30 boys and 32 girls) attending nursery schools took part in the baseline

study and were alternately allocated to conditions; their mean age was 4 years and 4 months, ranging from 3 years and 10 months to 4 years and 10 months. Prior to the study parents were contacted about the research, and none opted to exclude their children. Because some children did not attend all of the experimental sessions the final group sizes were: retrieval practice (15), re-presentation (15), and massed elaboration (13).

Materials

Three sets of seven plush toy animals were used. Each toy was assigned a normal, human name. As in Experiment 1, none of the toys represented well-known characters. One set was used to obtain a baseline measure. The other two sets were used in the experimental sessions, with the sets counterbalanced across study sessions.

Procedure

General. All testing took place in a separate, quiet area, and children were tested individually. At the end of each session they were praised for their efforts.

Baseline session. Each child participated in a baseline session during the week prior to participating in an experimental session. Before the session, seven plush toys were lined up on a table. The child and the experimenter sat with toys just outside the child's reach. Working from left to right, the experimenter named each toy and asked the child to name it. If the child did not name the toy, the experimenter continued to prompt the child until the child audibly and correctly named the toy; two children seemed unable to reliably repeat the names of the toys so the experimenter determined that their data would be excluded from the study. Children whose data were excluded were nevertheless included in all sessions (albeit with shorter, less demanding activities) to avoid any sense of failure or exclusion.

After all of the toys had been named, roughly one minute was spent in conversation with the child, after which the child's memory for the names was tested. A standard test procedure was followed throughout the experiment. The experimenter asked the child to name all of the toys that he or she could remember. The experimenter then specifically asked the child to name each toy that had not been spontaneously named. During this test procedure the experimenter did not provide the names for any of the toys.

First experimental session. Children were alternately assigned to one of the three study methods. One set of plush toys was lined up on a table. Each study segment lasted about 5 minutes.

For expanding retrieval practice, the procedure followed the schedule described in Table 3. For each event, the experimenter and the child each said the name of the toy once. *Introduction* involved the experimenter naming the toy and the child repeating the name. A *revisit* involved asking the child to say the name of the toy. If the toy was correctly named, the experimenter enthusiastically agreed and repeated the name; if not, the experimenter provided the name and the child repeated it. This procedure encouraged the child to practise recalling the names and scheduled the practice to maximize recall performance.

For the expanding re-presentation method, the procedure also followed the schedule described in Table 3 except that a *revisit* merely involved the experimenter providing the name and the child repeating it; the child was not encouraged to recall the name during the study segment. Note that in this study method, as in the expanding retrieval practice method, there are five events for each toy, and each event involved both the experimenter and the child saying the name once.

For the massed elaboration method, roughly 1 minute was spent on the first toy and 40 seconds on each toy thereafter. The experimenter and the child discussed each toy in turn with the experimenter first naming the toy and the child repeating the name. During the discussion, the experimenter named the toy a further four times and elicited the child's naming of the toy a further four times. This procedure matched the previous procedures on the time spent with each toy and the number of times the toy was named by both experimenter and child. The key

Table 3. *The schedule followed in the expanding retrieval practice and re-presentation conditions of Experiment 2*

	Event sequence																																				
Toy	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*16*	*17*	*18*	*19*	*20*	*21*	*22*	*23*	*24*	*25*	*26*	*27*	*28*	*29*	*30*	*31*	*32*	*33*	*34*	*35*	*36*	*37*	*38*	*39*
Filler toy	I		R					R			R											R						R							R		X	R	
Toy1		I		R		R				R					R																							
Toy2					I		R		R				R					R																				
Toy3												I		R		R				R				R														
Toy4																	I		R		R					R						R						
Toy5																							I		R		R			R				R				
Toy6																													I		R		R			R			R

Note: I = introduction of a toy's name. R = a revisit, retrieval, or re-presentation, depending upon the condition. X = brief, unrelated discussion between experimenter and child, included to create a short delay.

difference was that the study period for each toy was massed.

Following the study segment, the experimenter engaged the child in conversation for about 1 minute and then tested the child's memory for the names using the standard test procedure.

Second experimental session. To obtain delayed recall performance, the set of toys used in the first session were lined up on a table at the start of the session. Children's recall of the names was tested, using the standard procedure. Then, those toys were put away, and the remaining set of toys was lined up on the table. The study segment proceeded according to the method used in the first experimental session. Following the study segment, the children were again engaged in conversation for roughly one minute and were then tested on the second set of toys using the standard procedure.

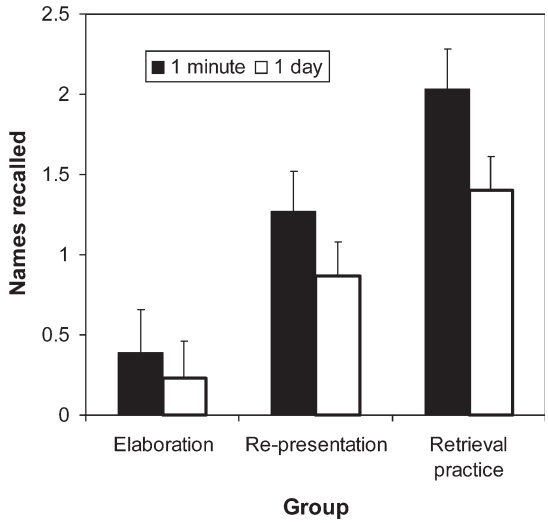


Figure 3. *Experiment 2: Mean recall of the names of the toys by children in the three conditions following delays of 1 minute and 1 day. Error bars indicate* +*1* SE.

Final session. To obtain delayed recall performance, the set of toys used in the second session were lined up on a table at the start of the session. Children's recall of the names was tested, using the standard procedure. Following this test, those toys were put away, and the set of toys used in the first session were lined up on the table. Children's recall of these names was tested again, using the standard procedure, to provide a measure of performance at a longer (2-day) delay.

Results

Strict scoring of the recall of the names was adopted so that the name had to be fully correct. More lenient scoring was examined and produced similar patterns but was harder to score reliably. Occasionally, children recalled a studied name, assigning it to the wrong toy; these intralist intrusions were scored as errors. Baseline performance was analysed first, and an ANOVA indicated that there were no significant differences in the preexperimental ability of the three groups, $F(2, 59) = 0.78$. Recall from the 1-minute and 1-day tests following the experimental study sessions is illustrated in Figure 3. A mixed ANOVA was carried out on the recall of the names in the experimental sessions with study method (massed elaboration, expanding re-presentation, and expanding retrieval practice) as the between-participants factor and delay (1 minute and 1 day) and toy set (Set 1 and Set 2) as within-participant factors. There were significant effects of study method, $F(2, 40) = 9.43$, $MSE = 2.93$, $p < .01$, and delay, $F(1, 40) = 19.82$, $MSE = 0.28$, $p < .01$; the effect of set was not significant, $F(1, 40) = 0.11$ (Set 1, $M = 1.05$; Set 2, $M = 1.01$), nor were any of the interactions significant. A planned comparison between retrieval practice and re-presentation revealed a significant benefit for retrieval practice, $F(1, 40) = 4.32$, $MSE = 2.93$, $p < .05$, $d = 0.9$. Similarly, expanding re-presentation led to significantly better recall than did massed elaboration, $F(1, 40) = 5.47$, $MSE = 2.93$, $p < .05$, $d = 1.0$. Cohen's d for the expanding retrieval practice versus massed elaboration comparison was 1.9. Figures 4 and 5 show the distributions for the recall of the names of the experimental toys, reduced by performance on the baseline test. These figures illustrate how individual children benefited from the different study methods.

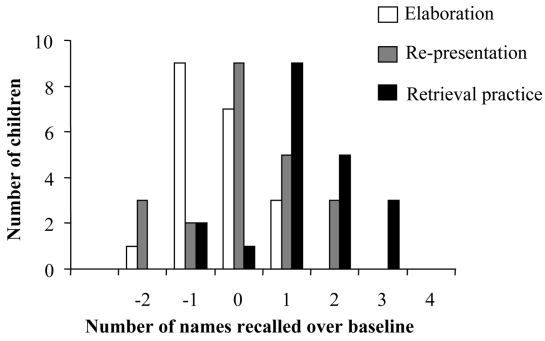


Figure 4. *Experiment 2, first set studied, test after 1-min: Distributions of children using the three study methods, showing the difference between their Day 1 experimental and baseline scores.*

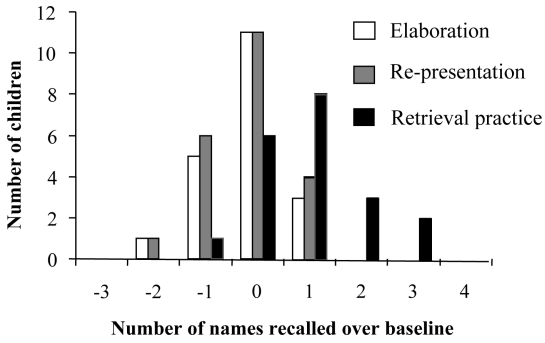


Figure 5. *Experiment 2, second set studied, test after 1-min: Distributions of children using the three study methods, showing the difference between their Day 2 experimental and baseline scores.*

Recall of the names of the toys used in the first experimental session was tested for a third time at the end of the third day. For this set of toys, recall performance on Day 2, following a 1-day delay, was as follows: massed elaboration, $M = 0.08$, $SD = 0.28$; expanding re-presentation, $M = 0.93$, $SD = 0.96$; expanding retrieval practice, $M = 1.47$, $SD = 1.19$. Recall performance on Day 3, following a further 24 hours, was as follows: massed elaboration, $M = 0.31$, $SD = 0.63$; expanding re-presentation, $M = 1.00$, $SD = 1.07$; expanding retrieval practice, $M = 1.60$, $SD = 1.12$. These data were analysed by ANOVA with study method (massed elaboration, expanding re-presentation, and expanding retrieval practice) as a between-participants factor and test delay (1 day or 2 days) as a within-participant factor. There was a significant effect of study condition, $F(2, 40) = 8.04$, $MSE = 1.56$, $p < .01$, but there was neither a significant effect of delay, $F(1, 40) = 1.86$, $MSE = 0.24$, nor a significant interaction of study condition and delay, $F(2, 40) = 0.20$, $MSE = 0.05$. Planned comparisons among the study conditions showed a marginally non-significant benefit for retrieval practice over re-presentations, $F(1, 40) = 3.08$, $MSE = 1.56$, $p = .08$, $d = 0.6$, a significant benefit for retrieval practice over massed elaboration, $F(1, 40) = 16.04$, $MSE = 1.56$, $p < .001$, $d = 1.5$, and a significant benefit for expanding re-presentations over massed elaboration, $F(1, 40) = 5.35$, $MSE = 1.56$, $p = .03$, $d = 0.9$.

Discussion

The powerful effect of expanding retrieval practice in aiding name learning by preschool children is indicated by the very large effect sizes for the comparison between the retrieval practice and the massed elaboration conditions. These study methods were matched for the time that the children spent considering each animal and the number of times that they heard and spoke the name. The substantial difference between the study methods results from the two main components of the expanding retrieval practice technique: the expanding schedule and retrieval practice. Comparisons with the re-presentation condition help to identify the relative contributions of these two components. The means and the effect sizes of the comparisons among the three conditions suggest that roughly half of the benefit of expanding retrieval practice for the preschool children came from the expanding schedule of re-presentations and about half from retrieval practice.

The almost identical recall for the first and second set of toys that the children studied indicated that the method was equally effective when used on a second set, with no signs of proactive interference. Similarly, when the toys from the first experimental session were retested on the final day, there had been no significant

loss since the first day, allaying concerns of retroactive interference.

Retrieval practice occurs whenever someone remembers something, so it is worthwhile to note that when tested at the 1- and 2-day delays all of the study methods included some retrieval practice. Practice on the first test, just 1 minute after the study session, would have improved recall performance on the following day for all children, regardless of study method (e.g., Carrier & Pashler, 1992). On the 2-day delayed test, the study methods being compared were actually (a) massed elaboration with practice after 1 minute and 1 day, (b) expanding re-presentation with practice after 1 minute and 1 day, and (c) expanding retrieval practice with five practice events on the first day and one on the second. The results in the final test have been influenced by these different patterns of retrieval practice. The lack of forgetting between Day 2 and Day 3 for all study methods can be attributed, at least in part, to the benefits of the retrieval practice that occurred for all methods. (Although the massed elaboration group was at floor on Day 2, the expanding re-presentation group was not; retrieval practice in the two tests would have helped them to maintain their level of performance.)

GENERAL DISCUSSION

The two experiments reported here demonstrate that expanding retrieval practice can offer a powerful technique for teaching children as young as 3 years. We found that the children very much enjoyed the technique, willingly engaged with it, and were delighted at the opportunity to play it a second time. Keeping their attention on the task at hand also seemed easier with the retrieval practice technique than with the other study conditions. We would not expect an expanding retrieval practice technique to replace other ways that young children learn, but for specific knowledge that is important for young children to acquire quickly and reliably, the technique offers a quick, popular, practical, and powerful opportunity.

Another important aspect of these results concerns the source of the benefits from expanding retrieval practice. The benefits of both spacing and testing are well documented. (For review and discussion, see Dempster, 1996.) The data from Experiment 2 suggest that, when the time and frequency of exposure are controlled, roughly half of the advantage over massed elaboration arises from the schedule of events, and the other half derives from practising retrieving the information.

Clearly learning the names of toys is not an essential part of young children's education, but other simple sets of associated information are. Young children are asked to learn basic letters, letter sounds, words, numbers, names of animals, names of objects, and much more. This sort of information is easily adapted to this sort of process. In fact, any information that can be presented on flash cards or in question and answer form is suitable. The key is to present the information, test it right away, before it is forgotten, and continue to test it at increasing intervals, before the information is lost.

In these experiments, children were taught and practised the material in one-on-one sessions. This sort of individualized instruction could be a sensible way to implement the technique in preschool classes. If the children usually played in groups of 4 to 6, then we would recommend that after all children in a group have had a study session similar to the experimental procedure, it would be an appropriate time for each child to have a further practice test to maintain the new knowledge a bit better. A refresher test of this sort was found to be beneficial for university students (Morris & Fritz, 2002), where an extra practice test with feedback after 2 weeks led to memory performance after 8 weeks that was equivalent to early performance 30 minutes after the original practice session. Alternatively, it could be possible to develop ways to use the technique in small groups or whole classes, as long as each child engages in remembering the answers during practice tests. Another possibly fruitful approach would be to develop computer software, presenting and testing information in an almost game-like format.

Retrieval practice may offer benefits beyond improving memory for the studied material. Many of these children appeared to develop greater self-confidence and showed some amazement and pride in their ability to remember the names both during the practice sessions and on the criterion tests. The use of an expanding schedule can apparently help to build confidence by repeatedly eliciting successful performance, even after a noticeable delay. Although adults are typically less candid than young children, our anecdotal experience with students playing the Name Game, a retrieval-based technique with an expanding schedule (Morris & Fritz, 2000, 2002; Morris, Fritz, & Buck, 2004) is similar. Well-constructed schedules of retrieval practice could benefit learners through improving their motivation and their metacognitive skills; this hypothesis is a potentially rich area for further research.

The scheduling of events and the nature of those events—the practice tests—are externally manipulated. One reason for examining the effects of the technique with preschool children is to determine whether or not benefits from these manipulations can be obtained in the absence of other strategic, wilful processing. Adults and older children are clearly capable of strategic engagement with the material that they are learning. The experience of practice tests provides them with feedback about the effectiveness of their strategies and enables them to adapt their approach to learning so that it is more effective. It seems reasonable to suppose that much of the benefit of retrieval practice could derive from this sort of strategic activity during the practice session. Preschool children, though, are unsophisticated learners who are not adept at strategic approaches to remembering (e.g., Flavell et al., 1970; Perlmutter & Myers, 1979). Because these children also benefit substantially from retrieval practice, we can safely conclude that strategic engagement during practice is not necessary to reap benefits from retrieval practice. Clearly learning and metamemory development are likely to benefit from that sort of strategic engagement, but there are benefits to be had from retrieval practice pure and simple.

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